

CS 471 – Lesson 8 In-Class Worksheet (Instructor Key)

Adversarial and Stochastic Decision Making

Guided In-Class Worksheet (Instructor-Led)

Objective: Explain minimax decision-making and zero-sum interactions in competitive environments, and describe expectimax and decision-making within stochastic environments.

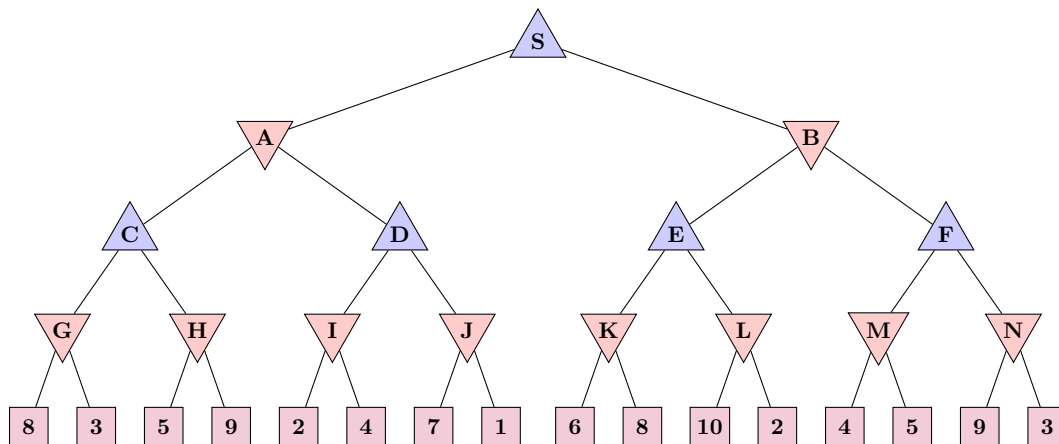
Symbology Legend:

- **Upward Triangle:** Max Agent Decision (You)
- **Downward Triangle:** Min Agent Decision (Adversary)
- **Circle:** Chance Node (Uncertain Outcomes)
- **Square:** Terminal Utility / State Value

Part 1: Adversarial Search (Minimax)

Scenario 1: As a cadet at the Air Force Academy, you are competing in a simulated zero-sum wargame. You are commanding a blue-force drone swarm (Max) attempting to maximize the number of enemy targets destroyed. An intelligent red-force commander (Min) is actively deploying countermeasures to minimize your success. This is a deterministic, zero-sum game.

Directions: Minimax values are computed recursively, working backwards from the terminal states. Trace the 5-layer minimax tree below to find the optimal policy against a rational adversary.



1. **Layer 4 & 3 Evaluation:** Calculate the values chosen by the Layer 4 Min nodes (G through N) and Layer 3 Max nodes (C through F).

Solution

Answer:

- **Min nodes (L4):** $G=3, H=5, I=2, J=1, K=6, L=2, M=4, N=3$.
- **Max nodes (L3):** $C=\max(3, 5) = 5; D=\max(2, 1) = 2; E=\max(6, 2) = 6; F=\max(4, 3) = 4$.

2. **Layer 2 Evaluation:** What value will Min Agent A choose? What value will Min Agent B choose?

Solution

Answer:

- Min Agent A minimizes between $C(5)$ and $D(2)$, choosing **2**.
- Min Agent B minimizes between $E(6)$ and $F(4)$, choosing **4**.

3. **Root Node & Policy:** What is the final minimax value at the root node (S), and which initial action should the Max Agent take?

Solution

Answer: The root node (S) maximizes between $A(2)$ and $B(4)$. The final value is **4**. The Max agent should take the path towards **Node B**.

4. **Theory:** Does Minimax model a worst-case or average-case perspective?

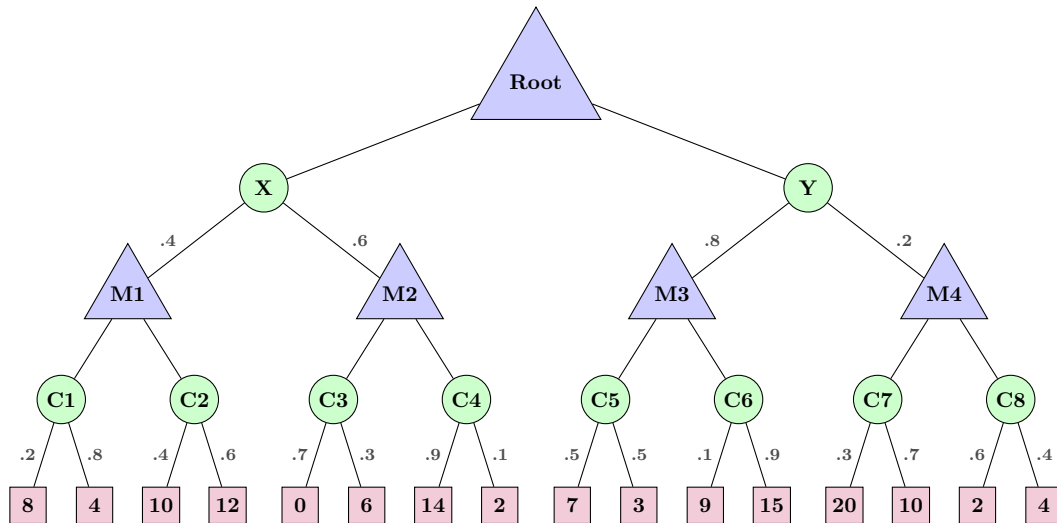
Solution

Answer: Minimax models the **worst-case perspective** because it assumes the opponent is a perfect player making optimal moves to minimize your score.

Part 2: Stochastic Search (Expectimax)

Scenario 2: Your blue-force drone is no longer facing an intelligent adversary. Instead, it must navigate the Rocky Mountains near the Academy in unpredictable weather conditions. The uncertain outcomes are controlled by chance (weather/sensor noise), not an adversary.

Directions: Calculate the expected utilities by taking the weighted average (expectation) of the children at each chance node. Ensure you apply the given probability weights on each branch.



5. **Layer 4 Expected Values:** Calculate the expected utility for chance nodes C1 through C8.

Solution

Answer: Values reflect average-case outcomes, calculated by $v = p_1(\text{val}_1) + p_2(\text{val}_2)$.

- **C1:** $0.2(8) + 0.8(4) = 1.6 + 3.2 = 4.8$
- **C2:** $0.4(10) + 0.6(12) = 4.0 + 7.2 = 11.2$
- **C3:** $0.7(0) + 0.3(6) = 0.0 + 1.8 = 1.8$
- **C4:** $0.9(14) + 0.1(2) = 12.6 + 0.2 = 12.8$
- **C5:** $0.5(7) + 0.5(3) = 3.5 + 1.5 = 5.0$
- **C6:** $0.1(9) + 0.9(15) = 0.9 + 13.5 = 14.4$
- **C7:** $0.3(20) + 0.7(10) = 6.0 + 7.0 = 13.0$
- **C8:** $0.6(2) + 0.4(4) = 1.2 + 1.6 = 2.8$

6. **Layer 3 Max Values:** What values will Max nodes M1, M2, M3, and M4 choose?

Solution

Answer: Max nodes choose the highest available expected utility from their children.

- **M1:** $\max(4.8, 11.2) = 11.2$

- **M2:** $\max(1.8, 12.8) = 12.8$

- **M3:** $\max(5.0, 14.4) = 14.4$

- **M4:** $\max(13.0, 2.8) = 13.0$

7. **Layer 2 Expected Values:** Calculate the expected utility for chance nodes X and Y.

Solution

Answer:

- **X:** $0.4(11.2) + 0.6(12.8) = 4.48 + 7.68 = 12.16$

- **Y:** $0.8(14.4) + 0.2(13.0) = 11.52 + 2.60 = 14.12$

8. **Root Node (Max):** What is the final expectimax value at the Root, and which path (towards X or Y) is optimal?

Solution

Answer: The final Expectimax value at the Root is $\max(12.16, 14.12) = 14.12$. The optimal policy for the blue-force drone is to take the path towards **Node Y**, as it yields the highest average-case outcome under optimal play.